

Building a Classification Model

Step 1: Calculate Initial Entropy

To measure the uncertainty of the dataset, we calculate its entropy based on the distribution of risk levels:

Risk Level	Count
High	6
Moderate	3
Low	5
Total	14

$$Entropy(S) = - \left(\frac{6}{14}\right) \log_2 \left(\frac{6}{14}\right) - \left(\frac{3}{14}\right) \log_2 \left(\frac{3}{14}\right) - \left(\frac{5}{14}\right) \log_2 \left(\frac{5}{14}\right)$$

Entropy(S) = 1.531

Step 2: Calculate Information Gain

Next, we are going to calculate the information gain which measures how much uncertainty is reduced by considering its features.

We are going to do that for each attribute:

Security

Security	Count	Class Distribution
Adequate	3	[0H, 1M, 2L]
None	11	[6H, 2M, 3L]

$$Entropy(S_{adequate}) = -(\frac{0}{3}) \times \log_2(\frac{0}{3}) - (\frac{1}{3}) \times \log_2(\frac{1}{3}) - (\frac{2}{3}) \times \log_2(\frac{2}{3})$$

$$Entropy(S_{adequate}) = 0.918$$

$$Entropy(S_{none}) = -(\frac{6}{11}) \times \log_2(\frac{6}{11}) - (\frac{2}{11}) \times \log_2(\frac{2}{11}) - (\frac{3}{11}) \times \log_2(\frac{3}{11})$$

$$Entropy(S_{none}) = 1.435$$

$$Gain(S, Security) = 1.531 - \left[\frac{3}{14} \times 0.918 + \frac{11}{14} \times 1.435 \right] = \boxed{0.206}$$

Salary

Salary Range	Count	Class Distribution
0–30K	4	[4H, 0M, 0L]
30K–60K	4	[2H, 2M, 0L]
Over \$60K	6	[0H, 1M, 5L]

$$Entropy(S_{0-30K}) = -(4/4) \times \log_2(4/4) = 0$$

$$Entropy(S_{30-60K}) = -(2/4) \times \log_2(2/4) - (2/4) \times \log_2(2/4) = 1.0$$

$$Entropy(S_{over60K}) = -(0/6) \times \log_2(0/6) - (1/6) \times \log_2(1/6) - (5/6) \times \log_2(5/6) = 0.650$$

$$Gain(S, Salary) = 1.531 - \left[\frac{4}{14} \times 0 + \frac{4}{14} \times 1.0 + \frac{6}{14} \times 0.650 \right] = \boxed{0.965}$$

Debt

Debt	Count	Class Distribution
High	7	[3H, 2M, 2L]

Debt	Count	Class Distribution
Low	7	[3H, 1M, 3L]

$$Entropy(S_{high}) = -(3/7) \times \log_2(3/7) - (2/7) \times \log_2(2/7) - (2/7) \times \log_2(2/7) = 1.556$$

$$Entropy(S_{low}) = -(3/7) \times \log_2(3/7) - (1/7) \times \log_2(1/7) - (3/7) \times \log_2(3/7) = 1.449$$

$$Gain(S, Debt) = 1.531 - \left[\frac{7}{14} \times 1.556 + \frac{7}{14} \times 1.449 \right] = \boxed{0.027}$$

Reputation

Reputation	Count	Class Distribution
Bad	3	[3H, 0M, 0L]
Unknown	5	[2H, 1M, 2L]
Good	6	[1H, 2M, 3L]

$$Entropy(S_{bad}) = 0$$

$$Entropy(S_{unknown}) = -(2/5) \times \log_2(2/5) - (1/5) \times \log_2(1/5) - (2/5) \times \log_2(2/5) = 1.522$$

$$Entropy(S_{good}) = -(1/6) \times \log_2(1/6) - (2/6) \times \log_2(2/6) - (3/6) \times \log_2(3/6) = 1.459$$

$$Gain(S, Reputation) = 1.531 - \left[\frac{3}{14} \times 0 + \frac{5}{14} \times 1.522 + \frac{6}{14} \times 1.459 \right] = \boxed{0.361}$$

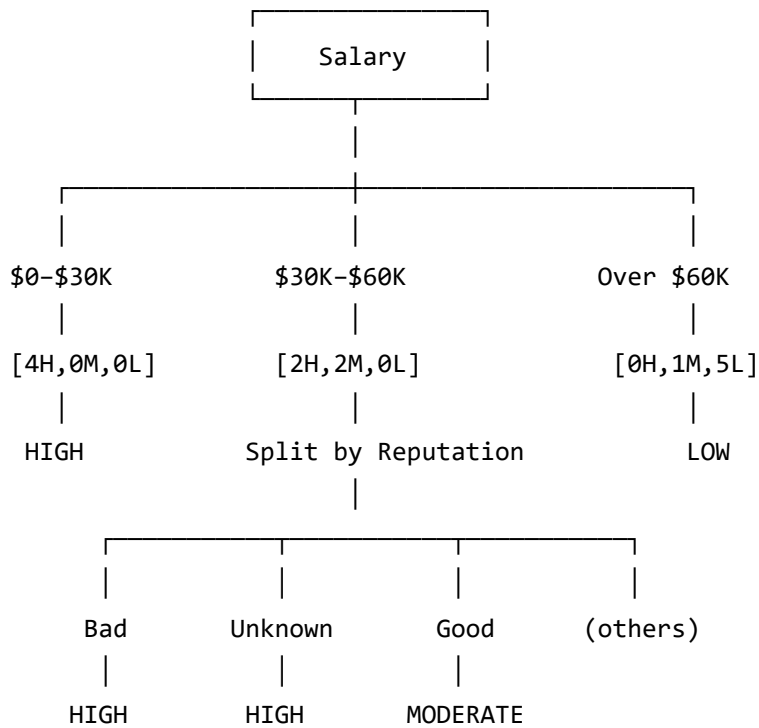
Summary of Information Gains

Attribute	Information Gain
Salary	0.965
Reputation	0.361
Security	0.206
Debt	0.027

We are going to build the decision tree with the root node being the highest information gain: **Salary**

Step 3: Build the Decision Tree

Root Node: Salary



Final Decision Rules

1. If **Salary = \$0-\$30K** → **High Risk**
2. If **Salary = Over \$60K** → **Low Risk**
3. If **Salary = \$30K-\$60K**:
 - Reputation = Bad → High Risk
 - Reputation = Unknown → High Risk
 - Reputation = Good → Moderate Risk

Final Model Summary

- **Root Node:** Salary (Highest Information Gain: 0.965)
- **Second Split (for $30K-60K$):** Reputation
- **Leaf Nodes:** High, Moderate, Low
- **Entropy Reduction:** Significant after Salary split